

XSPARKS

Engineering Excellence Division

FULL STACK ENGINEER TRAINING PROGRAM

ADDITIONAL LEARNING RESOURCES

Curated Day-Wise Resource Guide

Companion Document to Version 3.0 Curriculum

Effective Date: January 2026

Classification: Internal Use Only

How to Use This Guide

This resource guide is a companion to the XSPARKS Full Stack Engineer Training Program . It provides curated, high-quality learning resources organized day-by-day to align with the 4-week curriculum. Each day includes official documentation, video tutorials, online courses, articles, and recommended books relevant to that day's topics.

Resources are categorized as follows:

-  Official Documentation — Primary references from framework and tool creators
-  Video Tutorials & Courses — YouTube series, Udemy, Coursera, and free platforms
-  Blogs & Articles — In-depth technical articles and guides
-  Books — Recommended reading for deeper understanding
-  Hands-On Practice — Interactive labs, playgrounds, and exercises

Trainees are encouraged to use official documentation as the primary reference and supplement with video courses and articles for visual and deeper learning.

WEEK 1: FOUNDATIONS & CORE DEVELOPMENT

DAY 1: FRONTEND OPTIMIZATION & CODE QUALITY

Week 1 — React Context API, Lazy Loading, Pagination, ESLint/Prettier

► Official Documentation

- **React Official Docs — Context API** — The definitive reference for createContext, useContext, and provider patterns in modern React 19+. <https://react.dev/reference/react/createContext>
- **React Docs — Passing Data Deeply with Context** — Step-by-step guide on when and how to use Context to avoid prop drilling. <https://react.dev/learn/passing-data-deeply-with-context>
- **React Docs — Lazy Loading (React.lazy & Suspense)** — Official guide for code-splitting components with React.lazy() and Suspense boundaries. <https://react.dev/reference/react/lazy>
- **Chrome DevTools — Lighthouse** — Official documentation for running Lighthouse performance audits. <https://developer.chrome.com/docs/lighthouse/overview>
- **ESLint Official Getting Started** — Setup guide for configuring ESLint in JavaScript/TypeScript projects. <https://eslint.org/docs/latest/use/getting-started>
- **Prettier Documentation** — Configuration reference for Prettier code formatter integration. <https://prettier.io/docs/en/index.html>

► Video Tutorials & Courses

- **React Context API Tutorial — Codevolution (YouTube)** — Beginner-friendly walkthrough of Context API with practical examples. https://www.youtube.com/watch?v=5LrDIWkK_Bc
- **React Performance Optimization — Jack Herrington (YouTube)** — Deep dive into React.memo, useMemo, useCallback, lazy loading, and profiling. <https://www.youtube.com/watch?v=uoJLJf9SzY>
- **Web Performance Fundamentals — Frontend Masters** — Comprehensive course on web vitals, Lighthouse, and performance optimization techniques. <https://frontendmasters.com/courses/web-perf/>

► Blogs & Articles

- **React Context API Explained with Examples — freeCodeCamp** — Complete tutorial covering Context API setup, custom hooks, and best practices comparison with Redux. <https://www.freecodecamp.org/news/react-context-api-explained-with-examples/>
- **How to Use React Context Effectively — Kent C. Dodds** — Advanced patterns for structuring context providers and avoiding common pitfalls. <https://kentcdodds.com/blog/how-to-use-react-context-effectively>
- **React Context Tutorial — LogRocket Blog** — Updated guide with React 19 patterns, use cases, and Redux vs Context comparison. <https://blog.logrocket.com/react-context-tutorial/>

► Books

- **Learning React, 2nd Edition — Alex Banks & Eve Porcello (O'Reilly)** — Covers modern React patterns including hooks, context, and Suspense for data fetching.
- **React Key Concepts — Maximilian Schwarzmüller (Packt)** — Concise guide to React fundamentals including state management and performance.

DAY 2: BACKEND FUNDAMENTALS — NODE.JS

Week 1 — Express.js, RESTful APIs, JWT Authentication, Swagger► **Official Documentation**

- **Node.js Official Documentation** — Complete API reference for Node.js core modules, event loop, and runtime. <https://nodejs.org/en/docs>
- **Express.js Official Guide** — Routing, middleware, error handling, and template engines reference. <https://expressjs.com/en/guide/routing.html>
- **Swagger/OpenAPI Specification** — Official guide for writing OpenAPI specs and auto-generating documentation. <https://swagger.io/docs/specification/about/>

► **Video Tutorials & Courses**

- **Node.js and Express.js Full Course — freeCodeCamp (YouTube)** — 8-hour free course covering Node.js fundamentals through building Express REST APIs. <https://www.youtube.com/watch?v=Oe421EPjeBE>
- **Express.js Crash Course — Traversy Media (YouTube)** — Quick 1.5-hour crash course on Express routing, middleware, and REST API building. <https://www.youtube.com/watch?v=CnH3kAXSrmo>

► **Blogs & Articles**

- **MDN Express/Node.js Tutorial** — Mozilla's multi-part tutorial building a complete Express application with database integration. https://developer.mozilla.org/en-US/docs/Learn_web_development/Extensions/Server-side/Express_Nodejs
- **Express.js Tutorial — GeeksforGeeks** — Structured tutorial covering Express from fundamentals to advanced middleware patterns. <https://www.geeksforgeeks.org/node-js/express-js/>
- **JWT Authentication Tutorial — DigitalOcean** — Step-by-step guide to implementing JWT auth with Express. <https://www.digitalocean.com/community/tutorials/nodejs-jwt-expressjs>

► **Books**

- **Node.js Design Patterns, 3rd Edition — Mario Casciaro (Packt)** — Industry-standard book on Node.js patterns including streams, async, and scalability.
- **Express in Action — Evan Hahn (Manning)** — Focused deep-dive into Express middleware architecture and API design.

DAY 3: BACKEND FUNDAMENTALS — PYTHON FASTAPI

Week 1 — FastAPI, Pydantic, Async Programming, OAuth2► **Official Documentation**

- **FastAPI Official Tutorial** — The gold-standard tutorial by the creator — covers all features with interactive examples. <https://fastapi.tiangolo.com/tutorial/>
- **Pydantic V2 Documentation** — Complete reference for data validation, serialization, and model configuration. <https://docs.pydantic.dev/latest/>
- **Python asyncio Documentation** — Official Python docs on asynchronous programming with async/await. <https://docs.python.org/3/library/asyncio.html>

► **Video Tutorials & Courses**

- **FastAPI Full Course for Beginners — freeCodeCamp (YouTube)** — Free 5+ hour course building a complete FastAPI application from scratch. <https://www.youtube.com/watch?v=tLKKmouUams>
- **Modern APIs with FastAPI — Talk Python Training** — Project-based course building and deploying real APIs with FastAPI and Pydantic. <https://training.talkpython.fm/courses/modern-fastapi-apis>

► Blogs & Articles

- **FastAPI Official Learn Page** — Curated learning path recommended by the framework creator himself. <https://fastapi.tiangolo.com/learn/>
- **Introduction to FastAPI — DataCamp** — Interactive course focused on HTTP operations, validation, and async patterns. <https://www.datacamp.com/courses/introduction-to-fastapi>

► Books

- **Building Data Science Applications with FastAPI — François Voron (Packt)** — Covers FastAPI from basics to production with data science and ML integration focus.
- **Fluent Python, 2nd Edition — Luciano Ramalho (O'Reilly)** — Deep Python knowledge essential for async programming, type hints, and data models.

DAY 4: VERSION CONTROL & TESTING FOUNDATIONS

Week 1 — Git Workflows, Playwright, Puppeteer, Test Automation

► Official Documentation

- **Git Official Documentation** — Complete reference for Git commands, workflows, and internals. <https://git-scm.com/doc>
- **Playwright Official Docs** — Getting started guide, API reference, and best practices for Playwright testing. <https://playwright.dev/docs/intro>
- **Playwright — Writing Tests** — Official guide on test structure, assertions, fixtures, and hooks. <https://playwright.dev/docs/writing-tests>

► Video Tutorials & Courses

- **Git and GitHub for Beginners — freeCodeCamp (YouTube)** — One-hour crash course covering Git fundamentals, branching, and GitHub collaboration. <https://www.youtube.com/watch?v=RGQj5yH7evk>
- **Introduction to Playwright — Test Automation University** — Free structured course with exercises covering Playwright fundamentals and Page Object Model. <https://testautomationu.applitools.com/playwright-intro/>

► Blogs & Articles

- **Atlassian Git Tutorials** — Excellent visual guides for Git workflows, branching strategies, and advanced commands. <https://www.atlassian.com/git/tutorials>
- **Playwright Tutorial — BrowserStack** — Comprehensive guide from setup to advanced capabilities including parallel testing and API validation. <https://www.browserstack.com/guide/playwright-tutorial>
- **Build Your First E2E Test — Microsoft Learn** — Free Microsoft module on Playwright project structure, test reports, and VS Code debugging. <https://learn.microsoft.com/en-us/training/modules/build-with-playwright/>

► Books

- **Pro Git, 2nd Edition — Scott Chacon (Free Online)** — The definitive free book on Git — covers everything from basics to Git internals. <https://git-scm.com/book/en/v2>

DAY 5: SYSTEM DESIGN & DATABASE DESIGN FUNDAMENTALS

Week 1 — Scalability, Normalization, Indexing, CAP Theorem

► Official Documentation

- **PostgreSQL Official Documentation** — Complete reference for PostgreSQL SQL syntax, indexing, and administration. <https://www.postgresql.org/docs/current/>
- **MongoDB Manual** — Official MongoDB documentation covering CRUD, aggregation, indexing, and schema design. <https://www.mongodb.com/docs/manual/>

► Video Tutorials & Courses

- **System Design for Beginners — Gaurav Sen (YouTube)** — Popular YouTube channel with clear explanations of load balancing, caching, sharding, and system design patterns. <https://www.youtube.com/c/GauravSensei>
- **System Design Primer — GitHub** — Comprehensive open-source guide to system design concepts with study plans and practice problems. <https://github.com/donnemartin/system-design-primer>
- **Database Engineering Course — freeCodeCamp (YouTube)** — In-depth free course on database internals, indexing, normalization, and ACID properties. <https://www.youtube.com/watch?v=4cWkVbC2bNE>

► Blogs & Articles

- **Database Design Fundamentals — Educative** — Excellent comparison of MongoDB vs PostgreSQL with architecture trade-offs. <https://www.educative.io/blog/mongodb-versus-postgresql-databases>
- **MongoDB Schema Design Best Practices — GeeksforGeeks** — Covers embedding, referencing, denormalization, indexing, and sharding strategies. <https://www.geeksforgeeks.org/mongodb/mongodb-schema-design-best-practices-and-techniques/>
- **Use The Index, Luke! — SQL Indexing Guide** — Free online book explaining SQL indexing and query optimization concepts. <https://use-the-index-luke.com/>

► Books

- **Designing Data-Intensive Applications — Martin Kleppmann (O'Reilly)** — The must-read book on distributed systems, databases, and system design fundamentals.
- **System Design Interview Vol. 1 & 2 — Alex Xu** — Practical system design walkthroughs covering URL shorteners, chat systems, rate limiters, and more.

WEEK 2: APIs, DATABASES & AUTHENTICATION

DAY 6: API PROTOCOLS — REST, GraphQL, WEBSOCKET

Week 2 — RESTful Design, GraphQL Queries/Mutations, Real-Time Communication

► Official Documentation

- **GraphQL Official Learn** — Official introduction to GraphQL concepts, schemas, queries, and mutations. <https://graphql.org/learn/>
- **Apollo GraphQL Tutorial** — Hands-on tutorials for building GraphQL servers and clients. <https://www.apollographql.com/tutorials/>
- **Socket.IO Documentation** — Complete guide for WebSocket-based real-time communication with Socket.IO. <https://socket.io/docs/v4/>

► Video Tutorials & Courses

- **GraphQL Crash Course — Traversy Media (YouTube)** — Quick introduction to GraphQL server setup with queries, mutations, and resolvers. <https://www.youtube.com/watch?v=BcLNfwF04Kw>
- **WebSockets in 100 Seconds — Fireship (YouTube)** — Fast-paced visual explainer of WebSocket protocol and real-time patterns. https://www.youtube.com/watch?v=1BfCnjr_Vjg
- **REST API Design Best Practices — freeCodeCamp (YouTube)** — Covers versioning, status codes, pagination, filtering, and HATEOAS patterns. <https://www.youtube.com/watch?v=-MTSQjw5DrM>

► Books

- **RESTful Web APIs — Leonard Richardson (O'Reilly)** — Deep dive into REST API design principles, hypermedia, and best practices.
- **Learning GraphQL — Eve Porcello & Alex Banks (O'Reilly)** — Practical introduction to GraphQL from basic queries to production applications.

DAY 7: DATABASE IMPLEMENTATION — SQL (POSTGRESQL)

Week 2 — Connection Pooling, JOINS, CTEs, Migrations, EXPLAIN ANALYZE

► Official Documentation

- **PostgreSQL Tutorial** — Official beginner-friendly tutorial covering SQL basics through advanced queries. <https://www.postgresql.org/docs/current/tutorial.html>
- **PostgreSQL EXPLAIN Documentation** — Reference for query plan analysis and performance optimization. <https://www.postgresql.org/docs/current/sql-explain.html>

► Video Tutorials & Courses

- **PostgreSQL Full Course — Amigoscode (YouTube)** — Comprehensive free PostgreSQL tutorial from installation to advanced queries. <https://www.youtube.com/watch?v=qw--VYLpxG4>
- **SQL and PostgreSQL — Stephen Grider (Udemy)** — Highly rated course covering SQL fundamentals, complex queries, and database design with PostgreSQL. <https://www.udemy.com/course/sql-and-postgresql/>

► Hands-On Practice

- **SQLBolt — Interactive SQL Lessons** — Free interactive SQL tutorial with progressive exercises covering JOINS, subqueries, and aggregations. <https://sqlbolt.com/>

- **PostgreSQL Exercises** — Free practice problems specifically for PostgreSQL with instant feedback. <https://pgexercises.com/>
- **Mode SQL Tutorial** — Practical SQL lessons using real-world datasets for analysis. <https://mode.com/sql-tutorial/>

DAY 8: DATABASE IMPLEMENTATION — NOSQL & VECTOR DATABASES

Week 2 — MongoDB, Weaviate, ChromaDB, Embeddings, Semantic Search

► Official Documentation

- **MongoDB University (Free Courses)** — Free official courses on MongoDB CRUD, aggregation, indexing, and Atlas. <https://learn.mongodb.com/>
- **ChromaDB Documentation** — Getting started guide for the open-source embedding database. <https://docs.trychroma.com/>
- **Weaviate Documentation** — Complete reference for Weaviate vector database setup, schemas, and search. <https://weaviate.io/developers/weaviate>

► Video Tutorials & Courses

- **MongoDB Crash Course — Traversy Media (YouTube)** — Quick introduction to MongoDB concepts, Atlas setup, and CRUD operations. <https://www.youtube.com/watch?v=-56x56UppqQ>
- **Vector Databases Explained — Fireship (YouTube)** — Clear visual explanation of embeddings, similarity search, and vector DB use cases. <https://www.youtube.com/watch?v=klTvEwg3oJ4>

► Books

- **MongoDB: The Definitive Guide, 3rd Ed. — Shannon Bradshaw (O'Reilly)** — Comprehensive reference for MongoDB architecture, querying, and administration.

DAY 9: CACHING MECHANISMS & MULTI-THREADING

Week 2 — Redis, CDN, Cache Invalidation, Worker Threads, async

► Official Documentation

- **Redis Official Documentation** — Complete reference for Redis commands, data structures, and caching patterns. <https://redis.io/docs/>
- **Node.js Worker Threads** — Official API reference for multi-threaded operations in Node.js. https://nodejs.org/api/worker_threads.html
- **Python asyncio Documentation** — Official guide to Python's asynchronous programming patterns. <https://docs.python.org/3/library/asyncio.html>

► Video Tutorials & Courses

- **Redis Crash Course — Traversy Media (YouTube)** — Practical introduction to Redis data types, caching strategies, and Pub/Sub. <https://www.youtube.com/watch?v=jgpVdJB2sKQ>
- **Caching Demystified — ByteByteGo (YouTube)** — System design perspective on caching layers, invalidation strategies, and CDN patterns. <https://www.youtube.com/watch?v=dGAgxozNWFE>

DAY 10: AUTHENTICATION & NETWORK PROTOCOLS

Week 2 — Auth0, Firebase Auth, SMTP/IMAP, SSL/TLS, CORS

► Official Documentation

- **Auth0 Developer Documentation** — Complete guide for implementing Auth0 with RBAC, social login, and SSO. <https://auth0.com/docs>
- **Firebase Authentication Docs** — Setup guides for email, Google, GitHub, and other authentication providers. <https://firebase.google.com/docs/auth>
- **MDN HTTP/CORS Guide** — Definitive reference on Cross-Origin Resource Sharing mechanics. <https://developer.mozilla.org/en-US/docs/Web/HTTP/CORS>

► **Video Tutorials & Courses**

- **Auth0 in 100 Seconds — Fireship (YouTube)** — Quick overview of Auth0 concepts followed by practical implementation. <https://www.youtube.com/watch?v=yufqeJLP1rI>
- **Session vs Token Authentication — Fireship (YouTube)** — Clear comparison of session-based vs JWT token-based authentication. <https://www.youtube.com/watch?v=UBUNrFtuWo>
- **SSL/TLS Explained — ByteByteGo (YouTube)** — Visual explanation of the TLS handshake, certificates, and HTTPS. <https://www.youtube.com/watch?v=j9QmMEWmcfo>

WEEK 3: DEPLOYMENT, DEVOPS & DOCKER OPTIMIZATION

DAY 11: CLOUD DEPLOYMENT — AWS

Week 3 — EC2, RDS, S3, CloudFront, Lambda, VPC, IAM

► Official Documentation

- **AWS Documentation** — Official reference for all AWS services including EC2, RDS, S3, Lambda, and VPC. <https://docs.aws.amazon.com/>
- **AWS Well-Architected Framework** — Best practices for building secure, resilient, and cost-efficient cloud architectures. <https://aws.amazon.com/architecture/well-architected/>

► Video Tutorials & Courses

- **AWS Certified Cloud Practitioner — freeCodeCamp (YouTube)** — Free comprehensive course covering core AWS services and cloud concepts. <https://www.youtube.com/watch?v=SOTamWNgDKc>

► Hands-On Practice

- **AWS Free Tier** — 12-month free tier with EC2, S3, RDS, Lambda, and other services for hands-on practice. <https://aws.amazon.com/free/>
- **AWS Skill Builder** — Free official AWS training platform with labs and learning paths. <https://skillbuilder.aws/>

DAY 12: CLOUD DEPLOYMENT — GCP & SERVERLESS PLATFORMS

Week 3 — Google Cloud, Vercel, Render, Netlify, DigitalOcean

► Official Documentation

- **Google Cloud Documentation** — Official guides for Compute Engine, Cloud SQL, Cloud Run, and other GCP services. <https://cloud.google.com/docs>
- **Vercel Documentation** — Complete guide for deploying frontend and full-stack applications on Vercel. <https://vercel.com/docs>
- **Render Documentation** — Guides for deploying web services, databases, and background workers. <https://docs.render.com/>

► Video Tutorials & Courses

- **GCP Fundamentals — Google Cloud (YouTube)** — Official Google Cloud channel with tutorials, architecture discussions, and demos. <https://www.youtube.com/googlecloudtech>
- **Vercel & Next.js Deployment — Vercel (YouTube)** — Official guides for deploying and optimizing applications on Vercel. <https://www.youtube.com/c/VercelHQ>

DAY 13: DOCKER & CONTAINER OPTIMIZATION

Week 3 — Dockerfiles, Multi-Stage Builds, Image Optimization, Security

► Official Documentation

- **Docker Official Documentation** — Complete reference for Dockerfiles, Compose, networking, and volumes. <https://docs.docker.com/>

- **Docker Best Practices Guide** — Official guide for writing efficient, secure, and maintainable Dockerfiles. <https://docs.docker.com/build/building/best-practices/>

► Video Tutorials & Courses

- **Docker Containers and Kubernetes Fundamentals — freeCodeCamp (YouTube)** — Free hands-on course covering Docker CLI, images, Compose, and Kubernetes basics. <https://www.freecodecamp.org/news/learn-docker-and-kubernetes-hands-on-course/>
- **Docker in 100 Seconds — Fireship (YouTube)** — Quick visual overview of Docker concepts followed by practical demo. <https://www.youtube.com/watch?v=Gjnup-PuquQ>

► Books

- **Docker Deep Dive — Nigel Poulton** — The most recommended Docker book — covers images, containers, networking, and orchestration.

DAY 14: KUBERNETES & CONTAINER ORCHESTRATION

Week 3 — K8s Architecture, Deployments, Services, HPA, ConfigMaps

► Official Documentation

- **Kubernetes Official Tutorials** — Interactive tutorials for deploying apps, exploring clusters, and exposing services. <https://kubernetes.io/docs/tutorials/>
- **Kubernetes Basics Tutorial** — Step-by-step guide to deploying your first app on Kubernetes. <https://kubernetes.io/docs/tutorials/kubernetes-basics/>

► Video Tutorials & Courses

- **Kubernetes Course — TechWorld with Nana (YouTube)** — Highly recommended 4-hour free course covering all K8s fundamentals with practical demos. <https://www.youtube.com/watch?v=X48VuDVv0do>
- **Docker and Kubernetes: The Complete Guide — Stephen Grider (Udemy)** — Comprehensive course covering Docker through production Kubernetes deployment with CI/CD. <https://www.udemy.com/course/docker-and-kubernetes-the-complete-guide/>
- **KodeKloud Kubernetes for Beginners** — Interactive course with built-in labs for hands-on Kubernetes practice. <https://kodekloud.com/courses/kubernetes-for-the-absolute-beginners/>

► Hands-On Practice

- **Kubernetes Interactive Tutorial** — Official browser-based interactive tutorial requiring no local setup. <https://kubernetes.io/docs/tutorials/kubernetes-basics/>
- **Play with Kubernetes** — Free 4-hour Kubernetes playground for hands-on experimentation. <https://labs.play-with-k8s.com/>

► Books

- **Kubernetes Up and Running, 3rd Ed. — Brendan Burns et al. (O'Reilly)** — The standard reference book for Kubernetes concepts, patterns, and production deployment.

DAY 15: CI/CD PIPELINES

Week 3 — Jenkins, GitHub Actions, Test Automation, Deployment Rollbacks

► Official Documentation

- **GitHub Actions Documentation** — Complete guide for creating workflows, managing secrets, and automation. <https://docs.github.com/en/actions>

- **Jenkins User Documentation** — Official Jenkins docs covering pipeline syntax, plugins, and administration. <https://www.jenkins.io/doc/>

► **Video Tutorials & Courses**

- **GitHub Actions Tutorial — TechWorld with Nana (YouTube)** — Complete tutorial on GitHub Actions workflows for CI/CD automation. https://www.youtube.com/watch?v=R8_veQiYBjI
- **Jenkins Full Course — freeCodeCamp (YouTube)** — Free comprehensive course on Jenkins setup, pipelines, and integration. <https://www.youtube.com/watch?v=f4idgaq2VqA>

► **Blogs & Articles**

- **CI/CD Best Practices — GitLab** — Industry best practices for pipeline design, testing stages, and deployment strategies. <https://about.gitlab.com/topics/ci-cd/cicd-best-practices/>

WEEK 4: ADVANCED AI/ML & INFRASTRUCTURE

DAY 16: INFRASTRUCTURE AS CODE & OBSERVABILITY

Week 4 — Terraform, AWS CDK, Prometheus, Grafana, Loki

► Official Documentation

- **Terraform Official Tutorials** — Hands-on tutorials for infrastructure provisioning with Terraform. <https://developer.hashicorp.com/terraform/tutorials>
- **AWS CDK Developer Guide** — Complete reference for defining cloud infrastructure in code with AWS CDK. <https://docs.aws.amazon.com/cdk/v2/guide/home.html>
- **Prometheus Documentation** — Getting started with metrics collection, PromQL, and alerting. <https://prometheus.io/docs/introduction/overview/>
- **Grafana Documentation** — Setup guides for dashboards, data sources, and visualization. <https://grafana.com/docs/grafana/latest/>

► Video Tutorials & Courses

- **Terraform in 100 Seconds — Fireship (YouTube)** — Quick visual introduction to Terraform concepts and workflow. <https://www.youtube.com/watch?v=tomUWcQ0P3k>
- **Complete Terraform Course — freeCodeCamp (YouTube)** — Free 2.5-hour beginner-friendly Terraform course with AWS examples. <https://www.youtube.com/watch?v=7xngnjfllK4>
- **Prometheus & Grafana Tutorial — TechWorld with Nana (YouTube)** — Practical setup of monitoring stack with Prometheus, Grafana, and alerting. <https://www.youtube.com/watch?v=QoDqxm7ybLc>

DAY 17: LLM FUNDAMENTALS & MODEL TYPES

Week 4 — LLM Architectures, Thinking vs Non-Thinking Models, Quantization, Ollama

► Official Documentation

- **Ollama Documentation** — Setup guide for running LLMs locally with Ollama. <https://ollama.ai/>
- **Hugging Face Transformers Docs** — Comprehensive reference for working with transformer models. <https://huggingface.co/docs/transformers/>
- **Hugging Face Hub** — Browse and download thousands of pre-trained models for various tasks. <https://huggingface.co/models>

► Video Tutorials & Courses

- **Intro to Large Language Models — Andrej Karpathy (YouTube)** — 1-hour masterclass by former OpenAI researcher explaining LLM fundamentals from first principles. https://www.youtube.com/watch?v=zjkBMFhNj_g
- **Hugging Face NLP Course (Free)** — Free official course on transformer models, tokenization, and fine-tuning. <https://huggingface.co/learn/nlp-course/>

► Books

- **The LLM Engineer's Handbook — Paul Iusztin & Maxime Labonne** — Practical guide to structuring and deploying LLM applications in production.

- **Build a Large Language Model (From Scratch) — Sebastian Raschka** — Hands-on book building a GPT-like model from scratch for deep understanding.

DAY 18: PROMPT ENGINEERING & CONTEXT MANAGEMENT

Week 4 — Prompt Techniques, RAG Systems, DSPy, Context Optimization

► Official Documentation

- **Prompt Engineering Guide (DAIR.AI)** — The most comprehensive open-source guide to prompt engineering techniques and research. <https://www.promptingguide.ai/>
- **Anthropic Prompt Engineering Guide** — Official guide for effective prompting with Claude models. <https://docs.anthropic.com/en/docs/build-with-claude/prompt-engineering>
- **OpenAI Prompt Engineering Guide** — Official best practices for prompting GPT models. <https://platform.openai.com/docs/guides/prompt-engineering>
- **DSPy Documentation** — Framework for programmatic prompt optimization and LLM pipeline design. <https://dspy-docs.vercel.app/>

► Video Tutorials & Courses

- **ChatGPT Prompt Engineering for Developers — DeepLearning.AI (Free)** — Free short course by Andrew Ng and Isa Fulford covering systematic prompt techniques. <https://www.deeplearning.ai/short-courses/chatgpt-prompt-engineering-for-developers/>
- **LangChain for LLM Application Development — DeepLearning.AI (Free)** — Free course by Harrison Chase teaching the LangChain framework for building LLM apps. <https://www.deeplearning.ai/short-courses/langchain-for-llm-application-development/>
- **LangChain: Chat with Your Data — DeepLearning.AI (Free)** — Covers RAG fundamentals including document loading, splitting, retrieval, and answer generation. <https://www.deeplearning.ai/short-courses/langchain-chat-with-your-data/>
- **RAG Course — DeepLearning.AI on Coursera** — Full Coursera course covering RAG pipeline construction, vector databases, and evaluation. <https://www.coursera.org/learn/retrieval-augmented-generation-rag>

► Blogs & Articles

- **Prompt Engineering Guide — GitHub (DAIR.AI)** — Open-source repository with papers, techniques, notebooks, and model-specific guides. <https://github.com/dair-ai/Prompt-Engineering-Guide>

DAY 19: MULTI-AGENT SYSTEMS & AGENT MEMORY

Week 4 — A2A Protocol, Agent Memory, LangGraph, LangChain, LangSmith

► Official Documentation

- **LangChain Documentation** — Official docs for building LLM-powered applications with chains, agents, and tools. https://python.langchain.com/docs/get_started/introduction
- **LangGraph Documentation** — Framework for building stateful, multi-actor LLM applications with graph workflows. <https://langchain-ai.github.io/langgraph/>
- **LangSmith Documentation** — Platform for debugging, testing, and monitoring LLM applications. <https://docs.smith.langchain.com/>
- **Langfuse Documentation** — Open-source LLM observability platform for tracing, evaluation, and analytics. <https://langfuse.com/docs>

► Video Tutorials & Courses

- **Multi AI Agent Systems with crewAI — DeepLearning.AI (Free)** — Free course on building collaborative multi-agent workflows. <https://www.deeplearning.ai/short-courses/multi-ai-agent-systems-with-crewai/>
- **AI Agents in LangGraph — DeepLearning.AI (Free)** — Free course on building agentic AI applications with LangGraph. <https://www.deeplearning.ai/short-courses/ai-agents-in-langgraph/>
- **LangChain Full Course — freeCodeCamp (YouTube)** — Free comprehensive course on building LLM applications with LangChain. <https://www.youtube.com/watch?v=IG7Uxts9SXs>

► Books

- **Generative AI with LangChain — Ben Auffarth (Packt)** — Practical guide to building LLM applications including agents, chains, and RAG systems.

DAY 20: MULTIMODAL AI & MCP INTEGRATION

Week 4 — Text/Image/Audio/Video Processing, STT/TTS, MCP, Queue Systems

► Official Documentation

- **Model Context Protocol (MCP) Specification** — Official specification and guides for the MCP standard. <https://modelcontextprotocol.io/>
- **OpenAI Whisper** — Open-source speech recognition model for transcription and translation. <https://github.com/openai/whisper>
- **ElevenLabs Documentation** — API reference for text-to-speech generation. <https://elevenlabs.io/docs>
- **RabbitMQ Documentation** — Complete guides for message queuing, routing, and async processing. <https://www.rabbitmq.com/docs>

► Video Tutorials & Courses

- **MCP Explained — Anthropic (YouTube)** — Introduction to the Model Context Protocol and building MCP servers/clients. <https://www.youtube.com/watch?v=kQmXtrmQ5Zg>
- **Whisper Transcription Tutorial — AssemblyAI (YouTube)** — Practical guide to using OpenAI Whisper for speech-to-text in Python. https://www.youtube.com/watch?v=ABFqbY_rmEk
- **RabbitMQ Crash Course — Traversy Media (YouTube)** — Quick introduction to message queuing concepts and RabbitMQ patterns. <https://www.youtube.com/watch?v=NQ3fZtyXji0>

CAPSTONE PROJECT RESOURCES

The capstone project requires building a No-Code AI Model Fine-Tuning Platform. The following resources directly support capstone development:

Fine-Tuning & LoRA/QLoRA

- **PEFT Library Documentation** — Official documentation for Parameter-Efficient Fine-Tuning including LoRA and QLoRA. <https://huggingface.co/docs/peft/>
- **Fine-Tuning LLMs with Hugging Face — DeepLearning.AI (Free)** — Free course covering fine-tuning workflows, data preparation, and evaluation. <https://www.deeplearning.ai/short-courses/finetuning-large-language-models/>
- **QLoRA Paper Walkthrough — Yannic Kilcher (YouTube)** — Technical explanation of the QLoRA paper and memory-efficient fine-tuning. <https://www.youtube.com/watch?v=v9PHWGOa8HA>

Full-Stack Integration

- **React + FastAPI Full Stack — TestDriven.io** — Tutorial on building a full-stack app with React frontend and FastAPI backend. <https://testdriven.io/blog/fastapi-react/>
- **WebSocket with FastAPI — Official Docs** — Guide for implementing real-time WebSocket communication in FastAPI. <https://fastapi.tiangolo.com/advanced/websockets/>
- **Kaggle API Documentation** — Reference for programmatic dataset import from Kaggle. <https://github.com/Kaggle/kaggle-api>

Monitoring & Deployment

- **Prometheus + Grafana + Docker Tutorial — TechWorld with Nana** — Setting up production monitoring for containerized applications. <https://www.youtube.com/watch?v=QoDqxm7ybLc>
- **GitHub Actions for Docker and K8s — TechWorld with Nana** — CI/CD pipeline automation for containerized deployments. https://www.youtube.com/watch?v=R8_veQiYBjI

RECOMMENDED BOOKS — COMPLETE READING LIST

Frontend & JavaScript

- Learning React, 2nd Edition — Alex Banks & Eve Porcello (O'Reilly)
- JavaScript: The Good Parts — Douglas Crockford (O'Reilly)
- You Don't Know JS (Book Series) — Kyle Simpson (Free on GitHub)

Backend & APIs

- Node.js Design Patterns, 3rd Edition — Mario Casciaro (Packt)
- Fluent Python, 2nd Edition — Luciano Ramalho (O'Reilly)
- Building Data Science Applications with FastAPI — François Voron (Packt)
- RESTful Web APIs — Leonard Richardson (O'Reilly)

Databases & System Design

- Designing Data-Intensive Applications — Martin Kleppmann (O'Reilly) [MUST READ]
- System Design Interview Vol. 1 & 2 — Alex Xu
- MongoDB: The Definitive Guide, 3rd Ed. — Shannon Bradshaw (O'Reilly)
- SQL Performance Explained — Markus Winand

DevOps & Cloud

- Docker Deep Dive — Nigel Poulton
- Kubernetes Up and Running, 3rd Ed. — Brendan Burns et al. (O'Reilly)
- The Phoenix Project — Gene Kim, Kevin Behr, George Spafford
- Pro Git, 2nd Edition — Scott Chacon (Free: <https://git-scm.com/book/en/v2>)

AI/ML & LLMs

- The LLM Engineer's Handbook — Paul Iusztin & Maxime Labonne
- Build a Large Language Model (From Scratch) — Sebastian Raschka
- Generative AI with LangChain — Ben Auffarth (Packt)
- The Art of Prompt Engineering with ChatGPT — Nathan Hunter

FREE LEARNING PLATFORMS & TOOLS

Interactive Coding Platforms

- freeCodeCamp (<https://www.freecodecamp.org>) — Free full-stack curriculum with certificates
- The Odin Project (<https://www.theodinproject.com>) — Open-source full-stack curriculum
- SQLBolt (<https://sqlbolt.com>) — Interactive SQL tutorial with exercises
- PostgreSQL Exercises (<https://pgexercises.com>) — Hands-on PostgreSQL practice
- Play with Docker (<https://labs.play-with-docker.com>) — Browser-based Docker playground
- Play with Kubernetes (<https://labs.play-with-k8s.com>) — Browser-based K8s playground

Free Course Platforms

- DeepLearning.AI Short Courses (<https://www.deeplearning.ai/short-courses/>) — 60+ free AI/ML courses
- MongoDB University (<https://learn.mongodb.com>) — Free official MongoDB courses
- AWS Skill Builder (<https://skillbuilder.aws>) — Free AWS training and labs
- Microsoft Learn (<https://learn.microsoft.com>) — Free courses for Azure, .NET, and more
- Test Automation University (<https://testautomationu.applitools.com>) — Free testing courses
- KodeKloud (<https://kodekloud.com>) — DevOps and Kubernetes labs with free tier

YouTube Channels to Follow

- Fireship — Quick, visual tech explanations (Docker, React, AI in 100 seconds)
- TechWorld with Nana — DevOps, Docker, Kubernetes deep dives
- Traversy Media — Web development crash courses
- ByteByteGo — System design concepts and architecture
- freeCodeCamp — Full-length free courses on everything
- Jack Herrington — Advanced React patterns and performance
- Codevolution — React, Node.js, and TypeScript tutorials
- Andrej Karpathy — Deep learning and LLM fundamentals